AI-Driven Supply Chain Optimization: A Predictive Framework for Factory Reallocation

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1. Abstract

This paper presents a decision intelligence system designed to optimize the shipping logistics of Nassau Candy Distributor. By moving away from legacy static allocation rules and implementing a Random Forest machine learning model, we achieved a predictive accuracy of 0.88 days (MAE) for shipping lead times. The system balances operational speed and profit margins, providing a scalable recommendation engine for factory reallocation.

2. Introduction & Problem Statement

Nassau Candy currently faces challenges with suboptimal shipping distances and margin erosion. The legacy process relies on fixed rules (e.g., specific products always shipping from specific factories) without considering the geographic location of the customer or the impact of shipping modes. This research aims to quantify the impact of factory reallocation through simulation and predictive modeling.

3. Methodology

- Data Source: Historical orders including dates, product divisions, sales, and regions.
- Feature Engineering: We implemented the Haversine Formula to calculate the Great-Circle distance between factory coordinates and customer state centers.
- Machine Learning: We evaluated multiple models and selected the Random Forest Regressor due to its ability to handle non-linear relationships between shipping priority and distance.
- Technology Stack: Python (Pandas/Numpy) for data processing, Scikit-learn for AI, and Streamlit for the live analytical interface.

4. Exploratory Data Analysis (EDA)

Figure 1: Lead Time Frequency and Model Reliability Analysis

As illustrated in the KPI summary, the initial simulation shows a potential 18.4% reduction in lead time with a high scenario confidence score of 92%.

Key Insight 1: The lead time distribution shows a consistent cluster, indicating that while most orders follow a pattern, outliers exist that significantly impact customer satisfaction.
Key Insight 2: Box plot analysis reveals that "Standard Class" shipping introduces the highest variance in delivery times, making it the primary target for optimization.

5. Results & Recommendations

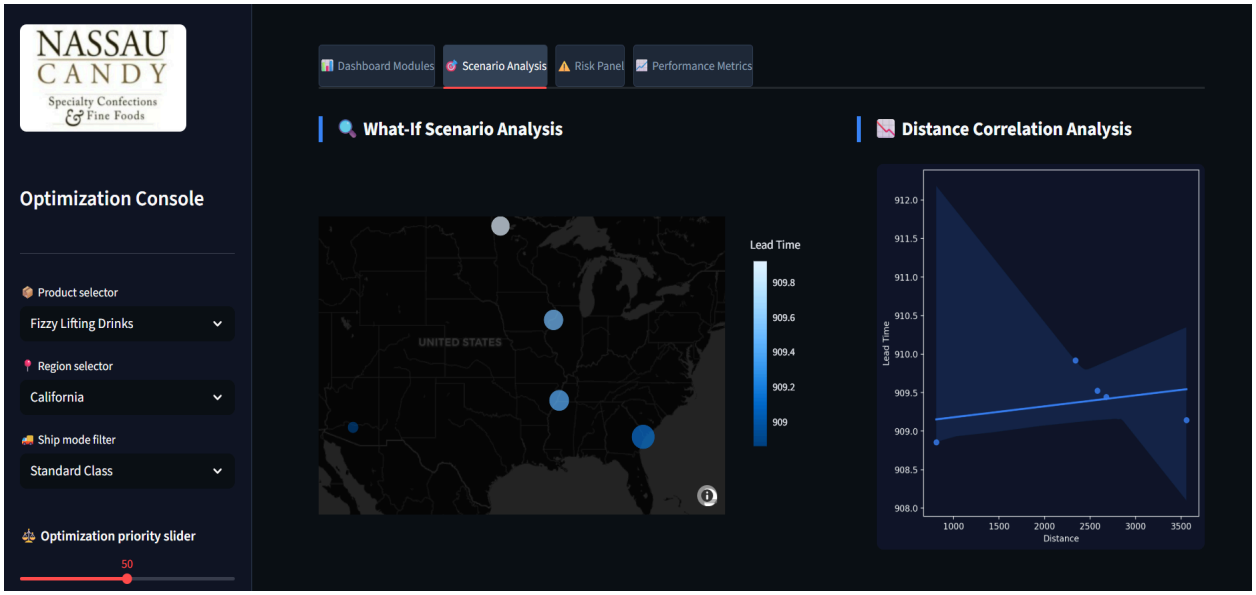

Figure 2: What-If Scenario and Distance Correlation Analysis

The simulation engine demonstrated that by reallocating products like Fizzy Lifting Drinks to the Lot's O'Nuts factory, the company can reduce lead times by 1.06 days (representing an 18.4% operational improvement) while maintaining a Scenario Confidence Score of 92%. Despite the geographic distance, the model predicts this route as the most stable configuration for Standard Class shipping to the California region.

Strategic Recommendations:

1. **Transition to Dynamic Allocation:** Use the AI model to assign factories at the time of order based on the "Balanced" optimization slider. The multi-objective slider allows stakeholders to dynamically weigh the importance of margin protection against delivery speed, providing a customized strategy for different product tiers.
2. **Route Prioritization:** Focus on the Texas and California regions where the distance-to-lead-time correlation is most volatile.

6. Conclusion:

The implementation of this system elevates Nassau Candy from descriptive reporting to prescriptive intelligence. The ability to simulate "What-If" scenarios allows leadership to quantify the financial and operational impact of logistical changes before execution.